

Codetantra Practical 5

Practice Lab Assignment

Course

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Essentials of Data Science
Laboratory - 2304102L - 2026

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5.1.1. Stacked Plot

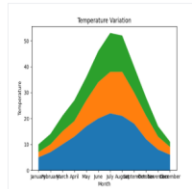
Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

Test case 1



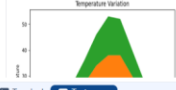
```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 data = {
5     'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July',
6             'August', 'September', 'October', 'November', 'December'],
7     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
8     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
9     'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
10 }
11 df=pd.DataFrame(data)
12 plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_B_Temperature'],df['City_C_Temperature'])
13 plt.xlabel('Month')
14 plt.ylabel('Temperature')
15 plt.title('Temperature Variation')
16 plt.show()
```

Average time: 0.456 s
Maximum time: 0.456 s
456.00 ms
456.00 ms
1 out of 1 shown test case(s) passed

Test case 1 456 ms Debug

Expected output

Actual output



Terminal Test cases

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Lab Assignment

Course

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Essentials of Data Science Laboratory (2024/25) - 2025

3.1.1 Titanic Dataset

Visualization: To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details: Write the code to create a series of visualizations as follows:
Bar Plot (Passenger Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).
- Use the color yellow for the bars.
- Title the plot as "Passenger Class Distribution".
- Label the x-axis as "Pclass" and the y-axis as "Count".

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use Legendbox for males and Legendbox for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as "Gender Distribution".

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passenger ages.
- Use Legendbox for the bars with black edges (`edgecolor='black'`).
- Set the number of bins to 10 for the histogram.
- Title the plot as "Age Distribution".

Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the `Survived` column.
- Use the colors Legendbox for survivors (1) and Legendbox for non-survivors (0).
- Title the plot as "Survival Count".
- Label the x-axis as "Survived (0 = No, 1 = Yes)" and the y-axis as "Count".

Scatter Plot (Fare vs Age):

- Create a scatter plot to visualize the relationship between the `Fare` and `Age` of passengers.
- Use orange for the data points.
- Title the plot as "Fare vs Age".
- Label the x-axis as "Age" and the y-axis as "Fare".

Note: Refer to the displayed plot in the sample test cases for better understanding.

Sample Test Cases

Test case 1

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Practical 3

Practice Lab Assignment

Numpy Array Operations

Lab Assignment

Numpy: Matrix Operations

Numpy: Horizontal and Vertical Stac...

Numpy: Custom Sequence Generation

Numpy: Arithmetic and Statistical Op...

Numpy: Copying and Viewing Arrays

Numpy: Searching, Sorting, Countin...

Student Data Analysis and Operations

Practical 4

Practice Lab Assignment

Pandas - series creation and manipu...

Dictionary to dataframe

Student Information

Lab Assignment

Practical 5

Practice Lab Assignment

Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use 30 bins for the histogram.
- Set the edge color of the bars to black (k).
- Label the x-axis as 'Age' and the y-axis as 'Frequency'.
- Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
passenger_id,survived,pclass,name,sex,age,sibsp parch,ticket,fare,cabin,embarked\n1,0,3,\"Braund, Mr. Owen Harris\",male,22,1,0,A/S 21371,7.25,S\n2,1,1,\"Cumings, Mrs. John Bradley (Florence Briggs Thayer)\",female,38,1,0,PC 17599,71.2833,C85,C\n3,1,1,\"Heikkinen, Miss. Laina\",female,26,0,B/STON/O2 3308282,7.925,S\n4,1,1,\"Hartwell, Mrs. Jacques Heath (Lily May Peel)\",female,35,1,0,S/30890,53.1,C123,S\n5,0,3,\"Allen, Mr. William Henry\",male,35,0,R/37450,8.65,S\n6,0,3,\"Moran, Mr. James\",male,0,0,338077,8.4383,Q\n7,0,1,\"McCarthy, Mr. Timothy J.\",male,34,0,0,17463,51.8625,C66,S\n8,0,3,\"Palsson, Master. Gosta Leonard\",male,21,1,349989,21.675,S\n9,1,3,\"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)\",female,27,0,2,347742,11.1333,S\n10,1,2,\"Nasser, Mrs. Nicholas (Adele Achen)\",female,14,1,0,237736,8.070C,
```

Note: Refer to the visible test case for better reference.

Sample Test CasesTest case 1

Histogra...

import pandas as pd\nimport matplotlib.pyplot as plt\n\ndata = pd.read_csv('Titanic-Dataset.csv')\n\n# Data Cleaning\ndata['Age'] = data['Age'].fillna(data['Age'].median(), inplace=True)\ndata['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)\ndata.drop('Cabin', axis=1, inplace=True)\n\n# Convert categorical features to numeric\data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})\ndata = pd.get_dummies(data, columns=['Embarked'], drop_first=True)\n\n# Write your code here for Histogramplt.hist(data['Age'],bins=30,edgecolor='k')plt.xlabel(\"Age\")plt.ylabel(\"Frequency\")plt.title("Age Distribution")plt.show()

Execution Time5.522 s522 msOptimization Time5.522 s522 ms1 out of 10 test case(s) passed

Age Distribution

Age Distribution

Terminal | Test cases

Course

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4.1 Practice Lab Assignment

4.1.1 Pandas - series creation and manipu...

4.1.2 Dictionary to dataframe

4.1.3 Student Information

4.2 Lab Assignment

4. Practical 5

5.1 Practice Lab Assignment

5.1.1 Stacked Plot

5.2 Lab Assignment

5.2.1 Titanic Dataset

5.2.2 Histogram of passenger information ...

5.2.3 Bar plot of survival rate of passengers

5.2.4 Bar Plot for Survival by Gender

5.2.5 Bar Plot for Survival by Pclass

5.2.6 Bar Plot for Survival by Embarked

5.2.7 Box plot for Age Distribution

5.2.8 Box Plot for Age by Survived

5.2.9 Box Plot for Fare by Pclass

5.2.10 Scatter Plot for Age vs. Fare

5.2.11 Scatter Plot for Age vs. Fare by Sur...

5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:
1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to 'bar'.
3. Add the title 'Survival Count' to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,7,25	5.42	NaN	S
2	1	1	Coltage, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71,2833	53.1	C85	C
3	1	3	McKimm, Miss. Lavinia	female	30	0	0	33082,7,936	5.0	NaN	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53,1,123	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8,45	8.45	NaN	S
6	0	3	Moran, Mr. James	male	19	0	0	538077,8,453	0	NaN	S
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463,51,8625	54.0	NaN	S
8	0	3	Paisson, Master. Gosta Leonard	male	2,1,1,349989	21,875	5	9,1,3	1,2	1,2	1,2
9	1	3	Tamoom, Mrs. Oscar W (Elizabeth Wilhelmine Berg)	female	27	0	2	347742,11,1333	5	NaN	S
10	1	2	Hassner, Mrs. Nicholas (Adele Achen)	female	34	1	0	237736,30,9708	0	NaN	C

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

Note: Refer to the visible test case for better reference.

Sample Test Cases

Test case 1

BarPlotOf...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 # Write your code here for Bar Plot for Survival Rate
16 sc = data['Survived'].value_counts()
17 sc.plot(kind='bar')
18 plt.title("Survival Count")
19 plt.xlabel("Survived")
20 plt.ylabel("Count")
21 plt.show()

Average time 0.611 s
Maximum time 0.611 s
1 out of 1 shown test case(s) passed

Expected output

Actual output

Survival Count

Survival Count

Terminal

Test cases

Prev Next

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4.1 Practice Lab Assignment

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5.1 Practice Lab Assignment

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5.2.1 Titanic Dataset

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5.2.3 Bar plot of survival rate of passengers

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5.2.8 Box Plot for Age by Survived

5.2.9 Box Plot for Fare by Pclass

5.2.10 Scatter Plot for Age vs. Fare

5.2.11 Scatter Plot for Age vs. Fare by Sur...

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:
1. Group the data by the "Sex" column, then use the value_counts() function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a stacked bar chart to display the survival counts.
3. Add the title "Survival by Gender" to the chart.
4. Label the x-axis as "Gender" and the y-axis as "Count".
5. The legend should indicate "Not Survived" and "Survived".

The Titanic dataset contains columns as shown below:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brandy, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Castle, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3308282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Alder, Mr. William Henry	male	30	0	0	374606	8.45	S	
6	0	3	Moran, Mr. James	male	0	0	0	338077	8.453	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	S46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	340809	21.075	S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	S	
10	1	2	Nasser, Mrs. Nicholas (Adele Achen)	female	34	1	0	237736	36.0708	C	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,Brandy, Mr. Owen Harris,male,22,1,0,A/5 21171,7.25,S
2,1,1,Castle, Mrs. John Bradley (Florence Briggs Thayer),female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,Heikkinen, Miss. Laina,female,26,0,0,STON/O2. 3308282,7.925,S
4,1,1,Futrelle, Mrs. Jacques Heath (Lily May Peel),female,35,1,0,113803,53.1,C123,S
5,0,3,Alder, Mr. William Henry,male,30,0,0,374606,8.45,S
6,0,3,Moran, Mr. James,male,0,0,0,338077,8.453,Q
7,0,1,McCarthy, Mr. Timothy J,male,54,0,0,17463,51.8625,S46,S
8,0,3,Palsson, Master. Gosta Leonard,male,2,3,1,340809,21.075,S
9,1,3,Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg),female,27,0,2,347742,11.1333,S
10,1,2,Nasser, Mrs. Nicholas (Adele Achen),female,34,1,0,237736,36.0708,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv("Titanic-Dataset.csv")
6
7# Data Cleaning
8data["Age"].fillna(data["Age"].median(), inplace=True)
9data["Embarked"].fillna(data["Embarked"].mode()[0], inplace=True)
10data.drop("Cabin", axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data["Sex"] = data["Sex"].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=["Embarked"], drop_first=True)
15sc = data.groupby("Sex")["Survived"].value_counts().unstack()
16sc.plot(kind="bar", stacked=True)
17plt.title("Survival by Gender")
18plt.xlabel("Gender")
19plt.ylabel("Count")
20plt.legend(["Not Survived", "Survived"])
21plt.show()

Average time: 0.732 s
Maximum time: 0.732 s
732.00 ms
732.00 ms
1 out of 1 shown test case(s) passed

Test Case 1
Expected output: Survival by Gender
Actual output: Survival by Gender

Terminal Test cases

Course

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4.1 Practice Lab Assignment

4.1.1 Pandas - series creation and manipu...

4.1.2 Dictionary to dataframe

4.1.3 Student Information

4.2 Lab Assignment

4.3 Practical 5

5.1 Practice Lab Assignment

5.1.1 Stacked Plot

5.2 Lab Assignment

5.2.1 Titanic Dataset

5.2.2 Histogram of passenger information ...

5.2.3 Bar plot of survival rate of passengers

5.2.4 Bar Plot for Survival by Gender

5.2.5 Bar Plot for Survival by Pclass

5.2.6 Bar Plot for Survival by Embarked

5.2.7 Box plot for Age Distribution

5.2.8 Box Plot for Age by Survived

5.2.9 Box Plot for Fare by Pclass

5.2.10 Scatter Plot for Age vs. Fare

5.2.11 Scatter Plot for Age vs. Fare by Sur...

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:
1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().
2. Use a stacked bar chart to display the survival counts.
3. Add the title "Survival by Pclass" to the chart.
4. Label the x-axis as "Pclass" and the y-axis as "Count".
5. The legend should indicate "Not Survived" and "Survived".

The Titanic dataset contains columns as shown below:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
3,0,3,"Brund, Mr. Owen Harris",male,21,1,0,0,5 2121,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
1,1,3,"Malkinen, Miss. Laina",female,28,0,0,STON/O2. 3308282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.45,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4533,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.0625,C46,S
8,0,1,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,1,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,36.4708,C

Note:

- Refer to the visible test case for better reference.
- Ensure you use the groupby() function with value_counts() to count the survivors and non-survivors for each Pclass.
- Do not manually create data or create a df without value_counts() as the value_counts() method for

BarPlotOf...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv("Titanic-Dataset.csv")
6
7 # Data Cleaning
8 data["Age"].fillna(data["Age"].median(), inplace=True)
9 data["Embarked"].fillna(data["Embarked"].mode()[0], inplace=True)
10 data.drop("Cabin", axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data["Sex"] = data["Sex"].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=["Embarked"], drop_first=True)
15 sc = data.groupby("Pclass")["Survived"].value_counts().unstack()
16 sc.plot(kind="bar", stacked=True)
17 plt.title("Survival by Pclass")
18 plt.xlabel("Pclass")
19 plt.ylabel("Count")
20 plt.legend(["Not Survived", "Survived"])
21
22 plt.show()

Average time 0.670 s Maximum time 0.670 s
670.00 ms 670.00 ms
1 out of 1 shown test case(s) passed

Test Case 1
Expected output
Actual output
Survival by Pclass
Survival by Pclass

Terminal Test cases

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5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.
The chart should display the following specifications:
1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked" to the chart.
4. Label the x-axis as "Embarked" and the y-axis as "Count".
5. Include a legend to distinguish between survivors and non-survivors (label the legend as "Survived" and "Not Survived").

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brand, Mr. Owen Harris	male	22	1	0	A/5 21271,7,29	5		S
2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71,2833	0	C	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7,925	5		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53-1,1123	5		S
5	0	3	Allan, Mr. William Henry	male	35	0	0	373469,8,0	5		S
6	0	3	Moran, Mr. James	male	0	0	0	330877,8,4583	0		Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463,51,8625	0		S
8	0	3	Paisson, Master. Gosta Leonard	male	2	1	0	369890,11,075	5		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)	female	27	0	2	347742,11,1333	5		S
10	1	2	Hassner, Mrs. Nicholas (Adelle Achen)	female	14	1	0	237736,10,8708	0		C

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Brand, Mr. Owen Harris",male,22,1,0,A/5 21271,7,29,S
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71,2833,C,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7,925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53-1,1123,S
5,0,3,"Allan, Mr. William Henry",male,35,0,0,373469,8,0,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8,4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51,8625,C,S
8,0,3,"Paisson, Master. Gosta Leonard",male,2,1,0,369890,11,075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)",female,27,0,2,347742,11,1333,S
10,1,2,"Hassner, Mrs. Nicholas (Adelle Achen)",female,14,1,0,237736,10,8708,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15sc = data.groupby('Embarked_Q')['Survived'].value_counts().unstack()
16sc.plot(kind='bar', stacked=True)
17plt.title('Survival by Embarked')
18plt.xlabel('Embarked')
19plt.ylabel('Count')
20plt.legend(['Not Survived', 'Survived'])
21plt.show()

Average time: 0.700 s
Maximum time: 0.700 s
1 out of 1 shown test case(s) passed

Test Case 1
Expected output: Survival by Embarked (stacked bar chart)
Actual output: Survival by Embarked (stacked bar chart)

Terminal

Test cases

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4.1 Practice Lab Assignment

4.2 Lab Assignment

4.2.1 Month with the Highest Total Sales

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4.2.5 Titanic Dataset Analysis and Data Cl...

4.2.6 Titanic Dataset Analysis and Data Cl...

4.2.7 Titanic Dataset Analysis and Data Cl...

4.2.8 Titanic Dataset Analysis and Data Cl...

5 Practical 5

5.1 Practice Lab Assignment

5.2 Lab Assignment

5.2.1 Titanic Dataset

5.2.2 Histogram of passenger information ...

5.2.3 Bar plot of survival rate of passengers

5.2.4 Bar Plot for Survival by Gender

5.2.5 Bar Plot for Survival by Pclass

5.2.6 Bar Plot for Survival by Embarked

5.2.7 Box plot for Age Distribution

5.2.8 Box Plot for Age by Survived

5.2.9 Box Plot for Fare by Pclass

5.2.10 Scatter Plot for Age vs. Fare

5.2.11 Scatter Plot for Age vs. Fare by Sur...

5.2.7. Box plot for Age Distribution

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brund, Mr. Owen Harris	male	22	1	0	1113771	7.25	0	0
2	1	1	Cottage, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	0	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2	53.1	0	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	0	0	113803	53.1	0	C
5	0	3	Aiken, Mr. William Henry	male	29	0	0	373499	53.1	0	S
6	0	3	Norman, Mr. James	male	27	0	0	10557	8.4543	0	0
7	0	3	McCarthy, Mr. Timothy J.	male	34	0	0	17463	51.8625	0	S
8	0	3	Palsson, Master. Gosta Leonard	male	27	1	1	349909	11.875	0	S
9	1	1	Townsend, Mrs. Oscar W (Ellisabeth Wilhelmine Berg)	female	37	0	0	147742	31.2833	0	S
10	1	3	Nasser, Mrs. Nicholas (Adele Achen)	female	34	1	0	237759	36.8786	0	C

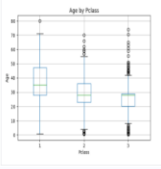
Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

Note: Refer to the visible test case for better reference.

Sample Test Cases

Test case 1

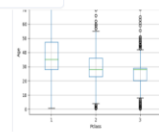
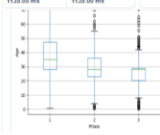


BoxPlot...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 # Write your code here for Box Plot for Age by Pclass
16 data.boxplot(column='Age', by='Pclass')
17 plt.suptitle('')
18 plt.title('Age by Pclass')
19 plt.xlabel('Pclass')
20 plt.ylabel('Age')
21 plt.show()
```

1.129 s 1.129 s 1.129 s

1 out of 1 shown test case(s) passed



Termin Test cases

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4.1. Practice Lab Assignment

4.2. Lab Assignment

4.2.1. Month with the Highest Total Sales

4.2.2. Best Selling Product

4.2.3. City that Sold the Most Products

4.2.4. Most Frequently Sold Product Pairs

4.2.5. Titanic Dataset Analysis and Data Cl...

4.2.6. Titanic Dataset Analysis and Data Cl...

4.2.7. Titanic Dataset Analysis and Data Cl...

4.2.8. Titanic Dataset Analysis and Data Cl...

4.2.9. Titanic Dataset Analysis and Data Cl...

4.2.10. Histogram of passenger information ...

4.2.11. Bar plot of survival rate of passengers

4.2.12. Bar Plot for Survival by Gender

4.2.13. Bar Plot for Survival by Pclass

4.2.14. Bar Plot for Survival by Embarked

4.2.15. Box plot for Age Distribution

4.2.16. Box Plot for Age by Survived

4.2.17. Box Plot for Fare by Pclass

4.2.18. Scatter Plot for Age vs. Fare

4.2.19. Scatter Plot for Age vs. Fare by Sur...

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:
1. Use the Survived column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to "Age by Survival".
3. Remove the default subtitle with plt.suptitle("").
4. Label the x-axis as 'Survived' and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Mr. Owen Harris",male,32,1,0,0,2101,7.25,S
2,0,1,"Capt. Wm. John Bradley",female,36,1,0,PC 17599,71.283,CB,C
3,1,3,"Mr. Heikkinen, Miss. Laina",female,26,0,0,STON/O, 3309142,7.93,S
4,1,1,"Mrs. Irene Heath (Lily May Peel)",female,35,1,0,310090,53.1,C33,S
5,0,1,"Alism, Mr. William Henry",male,18,0,0,37466,8.66,S
6,0,1,"Moran, Mr. James",male,0,0,310077,8.43,S
7,0,1,"McCarthy, Mr. Timothy J.",male,54,0,0,31743,53.805,64,S
8,0,1,"Passon, Master. Gosta Leonard",male,3,1,1,49900,11.87,S
9,1,3,"Johnson, Mrs. Oscar W (Ellenesth Vilhelmine Berg)",female,27,0,2,247402,11.539,S
10,1,3,"Nasser, Mrs. Nicholas (Nadine Achmed)",female,14,1,0,217756,16.876,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

Test case 1

Test case 1

Expected output

Actual output

Terminal

Test Cases

1 Pass 0 Fail 0 Skipped 0 Need 1

Course

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5.2.5. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **"Pclass"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	5	
2	1	1	Cutings, Mrs. John Bradley (Florence Briggs Tayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Hicks, Miss. Laina	female	26	0	0	STON/O2. 33091282	7.925	5	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	375369	8.05	5	
6	0	3	Moran, Mr. James	male	0	0	0	338075	5.40	5	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	646	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	1	1	349909	21.075	5	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)	female	27	0	1	347742	11.1333	5	
10	1	2	Masser, Mrs. Nicholas (Adele Achen)	female	14	1	0	237736	18.0788	C	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cutings, Mrs. John Bradley (Florence Briggs Tayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Hicks, Miss. Laina",female,26,0,0,STON/O2. 33091282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,375369,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,338075,5.40,S
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,646,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)",female,27,0,1,347742,11.1333,S
10,1,2,"Masser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,18.0788,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

Test case 1

Run by Pclass

BoxPlotFare

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15# Write your code here for Box Plot for Fare by Pclass
16data.boxplot(column='Fare', by = 'Pclass')
17plt.suptitle("")
18plt.title("Fare by Pclass")
19plt.xlabel("Pclass")
20plt.ylabel("Fare")
21plt.show()

Average time: 1.087 s
Maximum time: 1.087 s
1 out of 1 shown test case(s) passed

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5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:
1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **"Age"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	5	
2	1	1	Cunings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	McKinnon, Miss. Laine	female	26	0	0	STON/O2	53.00	5	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	5
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	5	
6	0	3	Moran, Mr. James	male	0	0	0	310077	0.4683	0	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	446	5
8	0	3	Palsson, Master. Gosta Leonard	male	2	1	1	349909	21.075	5	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	37	0	1	347742	11.1333	5	
10	1	2	Masser, Mrs. Nicholas (Adelie Achen)	female	14	1	0	237736	38.0788	C	

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,5
2,1,1,"Cunings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"McKinnon, Miss. Laine",female,26,0,0,STON/O2 53.00,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,5
6,0,3,"Moran, Mr. James",male,0,0,310077,0.4683,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,446,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,37,0,1,347742,11.1333,5
10,1,2,"Masser, Mrs. Nicholas (Adelie Achen)",female,14,1,0,237736,38.0788,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

Test case 1

Age vs. Fare

AgeFare5...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 # Write your code here for Box Plot for Fare by Pclass
16 plt.scatter(data['Age'], data['Fare'])
17 plt.suptitle("")
18 plt.xlabel("Age")
19 plt.ylabel("Fare")
20 plt.title("Age vs. Fare")
21 plt.show()

Average time: 0.902 s
Maximum time: 0.902 s
1 out of 1 shown test case(s) passed

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5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:
1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**) **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brand, Mr. Owen Harris	male	22	1	0	A/5 21271	7.25	S	
2	1	1	Cutings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Helakinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allan, Mr. William Henry	male	35	0	0	375469	8.49	S	
6	0	3	Moran, Mr. James	male	0	0	0	330877	8.4583	Q	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	1	1	349909	21.075	S	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg)	female	47	0	1	347742	11.1333	S	
10	1	2	Masser, Mrs. Nicholas (Adele Achen)	female	14	1	0	237736	39.8788	C	

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,Brand, Mr. Owen Harris, male, 22, 1, 0, A/5 21271, 7.25, S
2,1,1,Cutings, Mrs. John Bradley (Florence Briggs Thayer), female, 38, 1, 0, PC 17599, 71.2833, C85, C
3,1,3,Helakinen, Miss. Laina, female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4,1,1,Futrelle, Mrs. Jacques Heath (Lily May Peel), female, 35, 1, 0, 113803, 53.1, C123, S
5,0,3,Allan, Mr. William Henry, male, 35, 0, 0, 375469, 8.49, S
6,0,3, Moran, Mr. James, male, 0, 0, 0, 330877, 8.4583, Q
7,0,1,McCarthy, Mr. Timothy J, male, 54, 0, 0, 17463, 51.8625, E46, S
8,0,3,Palsson, Master. Gosta Leonard, male, 2, 1, 1, 349909, 21.075, S
9,1,3,Johnson, Mrs. Oscar W (Elisabeth Vilhelmine Berg), female, 47, 0, 1, 347742, 11.1333, S
10,1,2,Masser, Mrs. Nicholas (Adele Achen), female, 14, 1, 0, 237736, 39.8788, C

Note: Refer to the visible test case for better reference.

Sample Test Cases

Test case 1

AgeFare5...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 # Write your code here for Scatter Plot for Age vs. Fare by Survived
16 colours=data['Survived'].map({0: 'red', 1: 'blue'})
17 plt.scatter(data['Age'], data['Fare'], c=colours)
18 plt.title("Age vs. Fare by Survival")
19 plt.xlabel("Age")
20 plt.ylabel("Fare")
21 plt.show()

Average time: 0.071 s
Maximum time: 0.071 s
1 out of 1 shown test case(s) passed

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